

# CSE 232

## Computer Networks

### Homework-3

#### Group-15

|                |         |
|----------------|---------|
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#### DNS Messages

Design documentation that prints the DNS record printed by your modified **pcap** application, when you run `dig @198.41.0.4 www.example.com` from another terminal.

#### Source Code Implemented:

```
unsigned char* print_helper(unsigned char *start, unsigned char *msg_start, char
*name, size_t len)
{
    unsigned char *curr_pos = start, *return_pos = start;
    int is_compressed = 0;
    while((char)*curr_pos != 0){
        uint16_t offset = ntohs((__be16*)curr_pos);
        if ((offset >> 14) == 3){
            if (is_compressed == 0){
                return_pos = curr_pos + 2;
                is_compressed = 1;}
            curr_pos = msg_start + (offset & 0x3FFF);}

        else {
            int length = (int)*curr_pos;
            curr_pos++;
            for (int i = 0; i < length; i++)
                *name++ = (char)*curr_pos++;
            *name++ = '.';}}
    if (is_compressed == 0)
        return_pos = curr_pos + 1;
```

```
*name = '\0';  
return return_pos;  
}
```

## Output Query for the command

dig @198.41.0.4 www.example.com

[illegible]

